Assignment - 02

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Course :- CSE422

①

Answer no. 01

a

Elitism is an algorithm that replace the next generation's worst to the best of the generation. This algorithm helps to increase the convergence rate. For this reason it gets easier to reach the goal faster.

b

The single point crossover randomly selects a point from where it creates two offspring by exchanging the chromosome between the two parents from the same selected points.

$P_1 \longrightarrow 100|11001$
$P_2 \longrightarrow 001|01110$
$\Rightarrow$ $C_1 = 10001110$
$C_2 = 11001001$

The two point crossover is the process where instead of a point it randomly selects two point to exchange the chromosome & create offsprings from the parents

$P_1 \longrightarrow 100|11|001$
$P_2 \longrightarrow 001|01|110$
$\Rightarrow$ $C_1 = 10001001$
$C_2 = 00111110$

c

Some stopping condition that genetic algorithm can implement are given below:-

(i) After a certain number of generation the algorithm must be stopped.

(ii) If the threshold of a fitness level reached /exceed then algorithm must be stopped.

(iii) When the diversification of population stops / get fixed, then it should stop.

d

The importance of mutation is it will create new genes within the offspring chromosomes to create diversity

e

The importance of crossover is it will create a new offspring chromosome from the parent to create diversity

Answer no. 02

a

$P_1 \rightarrow$ abcdef

$P_2 \rightarrow$ afcbed

$P_3 \rightarrow$ adcfedb

$P_4 \rightarrow$ edafcb

fitnee function = Σ Path cost

b

$P_1 \rightarrow 10 + 19 + 22 + 5 + 12 = 68$

$P_2 \rightarrow 6 + \overset{15}{\cancel{10}} + 19 + 10 + 5 = 55$

$P_3 \rightarrow 8 + 22 + 15 + 12 + 10 = 67$

$P_4 \rightarrow 5 + 8 + 6 + 15 + 19 = 53$

Using the Roulette wheel selection —

$P_1 \rightarrow 68$

$P_2 \rightarrow 55$

$P_3 \rightarrow 67$

$P_4 \rightarrow 53$

243

$P_1 = 68/243 \times 100\% = 27.93\%$

$P_2 = 55/243 \times 100\% = 22.63\%$

$P_3 = 67/243 \times 100\% = 27.57\%$

$P_4 = 53/243 \times 100\% = 21.81\%$

P_2 & $P_4 \rightarrow$ best

c

Uniform Cross Over

$P_2 \to$ a f e b e d [1 = flip , 0 = don't flip].

$P_4 \to$ e d a f c b

MUX $\to$ $\boxed{0110110}$

$C_1 \to$ a d a b c d

$C_2 \to$ e f c f e b

Two point crossover

$P_2 \to$ a f | c b e | d

$P_4 \to$ e d | a f c | b

$C_3 \to$ a f a f c d

$C_4 \to$ e d c b e b

d

$C_1' \to$ a d e b c f ×

$C_2' \to$ e d b f a b ×

$C_3' \to$ c d e f a b ✓

$C_4' \to$ a e b f c d. ×

Only C_3' in eligible rest are not

(V)

Answer no. 03

a

$P_1 \rightarrow 1, 5, 7, 2, 3, 8, 4, 6$

$P_2 \rightarrow 5, 1, 2, 8, 6, 4, 3, 7$

$P_3 \rightarrow 6, 2, 3, 1, 7, 4, 3, 7$

$P_4 \rightarrow 2, 7, 1, 6, 5, 5, 3, 3$

fitness function $\Rightarrow$ Max No. of collision − Actual no. of collision

b

$P_1 \rightarrow {}^8C_2 - 4 = 24$

$P_2 \rightarrow {}^8C_2 - 5 = 23$

$P_3 \rightarrow {}^8C_2 - 3 = 25$

$P_4 \rightarrow {}^8C_2 - 6 = 22$

P_1 & P_3 are the selected chromosome.

(VI)

c

$P_1 \rightarrow 1, 5, 7, | 2, 3, 8, 4, 6$

$P_2 \rightarrow 6, 2, 3, | 1, 7, 4, 3, 7$

$C_1 \Rightarrow 1\ 5\ 7\ 1\ 7\ 4\ 3\ 7$

$C_2 \Rightarrow 6\ 2\ 3\ 2\ 3\ 8\ 4\ 6$

$P_1 \rightarrow 1\ 5\ 7\ 2 \mid 3\ 8\ 4\ 6$

$P_2 \rightarrow 6\ 2\ 3\ 1 \mid 7\ 4\ 3\ 7$

$C_3 \Rightarrow 1\ 5\ 7\ 2\ 7\ 4\ 3\ 7$

$C_4 \Rightarrow 6\ 2\ 3\ 1\ 3\ 8\ 4\ 6$

d

$C_1' \Rightarrow 1\ 5\ 1\ 7\ 2\ 4\ 3\ 7$

$C_2' \Rightarrow 6\ 2\ 3\ 2\ 4\ 8\ 6\ 1$

$C_3' \Rightarrow 1\ 5\ 2\ 7\ 8\ 4\ 3\ 1$

$C_4' \Rightarrow 6\ 2\ 3\ 1\ 3\ 8\ 4\ 7$

(VII)

e

$c_1' = 8c_2 - 4 = 24$

$c_2' = 8c_2 - \cancel{4}5 = \cancel{24}\ 23$

$c_3' = 8c_2 - 5 = 23$

$c_4' = 8c_2 - 5 = 23$

Answer no. ~~04~~ 05

a

Alpha beta pruning is a Depth First Search.

b

No, it doesn't as it will travell to every leaf node.

c

Yes, they will, but using alpha beta pruning will cost less time.

VIII

Answer no. 04